PRD: AI Internal Linking & Site Structure Agent

1. Problem Statement

Websites accumulate structural debt over time: orphaned pages, missing sitemap entries, authority trapped on low-value pages, and money pages that never receive the internal support their business importance warrants. Manually auditing and fixing this at scale is impractical — a naive pairwise comparison of all pages to find link opportunities is an $O(n^2)$ problem, and most site owners lack the graph/embedding tooling to do it properly.

This agent automates internal-link and site-structure optimization: it detects structural issues, understands the site's topical and authority graph, and produces prioritized, human-reviewable link suggestions — without ever requiring brute-force page-to-page comparison.

2. Goals

The system should:

1. **Catch simple structural issues cheaply** — orphan pages, pages missing from the sitemap, broken internal links — via fast rule-based checks before any expensive semantic work runs.
2. **Increase authority flowing to money pages** — specifically via links from informational pages, whether inside the same topic cluster or (more commonly, expected) across clusters via bridge links.
3. **Preserve a navigable, tree-like site structure** — pages within the same category/subcategory should cluster together and interlink coherently, mirroring the site's actual taxonomy.
4. **Avoid wasting authority on leaf/dead-end pages** — pages that can't redistribute equity further (thin, non-valuable, or structurally terminal) should not be link targets just because they're under-linked.
5. **Respect a per-page link budget** — never suggest pushing a page past its sustainable number of contextual links.
6. **Use simple deterministic rules as an early filter** — cheap, rule-based checks should narrow the candidate pool before any embedding/LLM work is spent on it.
7. **Weight authority by real performance data** — per-page CTR/impressions should factor into how

authority and priority are computed, not just raw link counts.

8. **Use SERP data to validate relatedness** — in addition to semantic similarity, check whether Google itself treats candidate pages as topically adjacent.

Underlying all of the above: the ultimate outcome is a better *user* experience — links that help real visitors reach what they need, not just links engineered for crawlers.

3. Non-Goals

- Automated content creation or rewriting.
- External backlink acquisition.
- Full technical crawl-budget management (log-file analysis, server-side crawl throttling) beyond basic orphan/sitemap detection.
- Fully autonomous publishing — all suggestions are reviewed by a human before going live.

4. Page Classification (prerequisite)

Every page is classified by type before entering the pipeline:

- **Money pages** — pages with direct commercial/conversion value (product, download, signup).
- **Informational pages** — blog/content pages that primarily earn traffic and rank for informational queries.
- **Utility/boilerplate pages** — legal, login, pagination, tag archives — excluded from being link targets regardless of authority deficit.
- **Leaf pages** — pages with low outgoing link counts that would act as authority dead-ends if linked to.

This classification gates every downstream stage — it's what stops the system from ever suggesting a link to a Terms of Service page just because it's under-linked.

5. Pipeline

Stage 1 — Cheap structural audit (rule-based, no ML).
Crawl the site, build the link graph, and flag: orphan pages (zero inlinks), pages missing from the sitemap, broken (404) internal links, and excessive crawl depth. This runs first and cheaply removes/flags issues before any semantic work is spent.

Stage 2 — Graph construction with editorial/template separation.
Build two link views: the full graph (all links, used for actual PageRank-style equity math) and a cleaned "editorial" graph with nav/footer/sidebar/breadcrumb links stripped out (used for hub detection, cluster discovery, and relevance signals, since template links appear identically everywhere and would create false structural signals).

Stage 3 — Authority scoring.
Compute a PageRank/InRank-style score per page from the full link graph, then weight/adjust it using real performance data — CTR and impressions from Search Console/Analytics — so that authority scoring reflects actual page performance and potential, not just link topology.

Stage 4 — Clustering and hub detection.
Cluster pages by topical embedding similarity (or existing category/subcategory taxonomy where available) to form the tree-like structure — same category/subcategory pages grouped together. Within each cluster, algorithmically detect the hub (a page that links to, and is linked back by, most other pages in the cluster) rather than requiring manual pillar tagging.

Stage 5 — Chunk-level candidate generation.
Rather than comparing whole pages, chunk page content and embed at the chunk level, so a narrow,

specific passage can be matched to an equally narrow passage elsewhere — this also yields the exact in-content location and context for the eventual anchor. Candidate retrieval uses an ANN index (e.g. FAISS/HNSW) so this scales sub-quadratically regardless of site size.

Stage 6 — SERP-based validation.
Cross-check top semantic candidates against SERP overlap — do the two pages actually compete or co-rank for related queries — as a second relevance filter on top of embedding similarity.

Stage 7 — Target prioritization and gating.
From pages that pass the page-type gate (money or valuable informational pages only — never utility/leaf pages), rank by authority deficit (adjusted for CTR/impressions from Stage 3) to decide which pages most need new inbound links this cycle.

Stage 8 — Bridge link detection.
Where a chunk-level match spans two different cluster hubs, flag it as a bridge-link candidate. Bias this detection toward the **informational-cluster → money-page-cluster** direction, since that's the direction that converts blog-earned authority into business value, and is expected to be the more common case.

Stage 9 — Structural gap detection.
Flag spokes whose only link back to their cluster hub is

a nav/template link with no contextual in-body backlink — the cleaned editorial graph would otherwise hide this as "no relationship" even though the architecture implies one.

Stage 10 — Link budget enforcement.
Compute each candidate source page's remaining contextual link budget (word-count-derived ceiling minus existing template and contextual links) and cap auto-approved suggestions per page at that ceiling. Excess candidates are deprioritized into a backlog rather than dumped.

Stage 11 — LLM review pass.
The full deterministic candidate list (all suggestion types: new links, bridge links, nav-gap fixes) is passed to an LLM to remove suggestions that are contextually wrong, redundant, or unnatural before anything reaches a human reviewer.

Stage 12 — Prioritized output.
Suggestions are scored and queued in a persistent backlog rather than freshly recomputed each run. Each run pulls a fixed **suggestion budget** (not a page count) off the top, so review volume stays predictable regardless of how many chunk-level matches any given page produces. Rejected suggestions are excluded from resurfacing.

6. Success Metrics (KPIs)

KPI	Metric
Structural health	Orphan pages, missing-sitemap pages, broken internal links trending to zero over time
Money page authority	Internal authority score (CTR-weighted) of designated money pages, tracked over time; count/ratio of info→money links added
Site tree coherence	% of pages correctly clustered with their category/subcategory hub; reduction in cross-taxonomy stray links
Authority efficiency	Reduction in links pointing to leaf/dead-end/utility pages
Budget compliance	Zero pages exceeding computed link budget after suggestions are applied
Suggestion precision	% of LLM-approved suggestions accepted by human reviewers (proxy for candidate quality)
Business outcome	Organic traffic/conversion movement on money pages after implementation (lagging indicator)

7. Open Questions / Risks

- How reliably can "money page" be auto-classified across different site types, vs. requiring manual tagging?
- Chunk boundary strategy (fixed-size vs. semantic/paragraph-based) will affect match quality — needs experimentation.
- Similarity/SERP-overlap thresholds will likely need per-site tuning rather than a single global constant.
- CTR/impressions data freshness and sparsity (new pages have none) needs a fallback scoring path.

8. Future Considerations

- Anchor text variation and optimization.
- Automated placement suggestions (not just "link here," but suggested phrasing).
- Periodic re-scoring cadence and drift detection as the site grows.
- Multi-language / multi-region site structures.